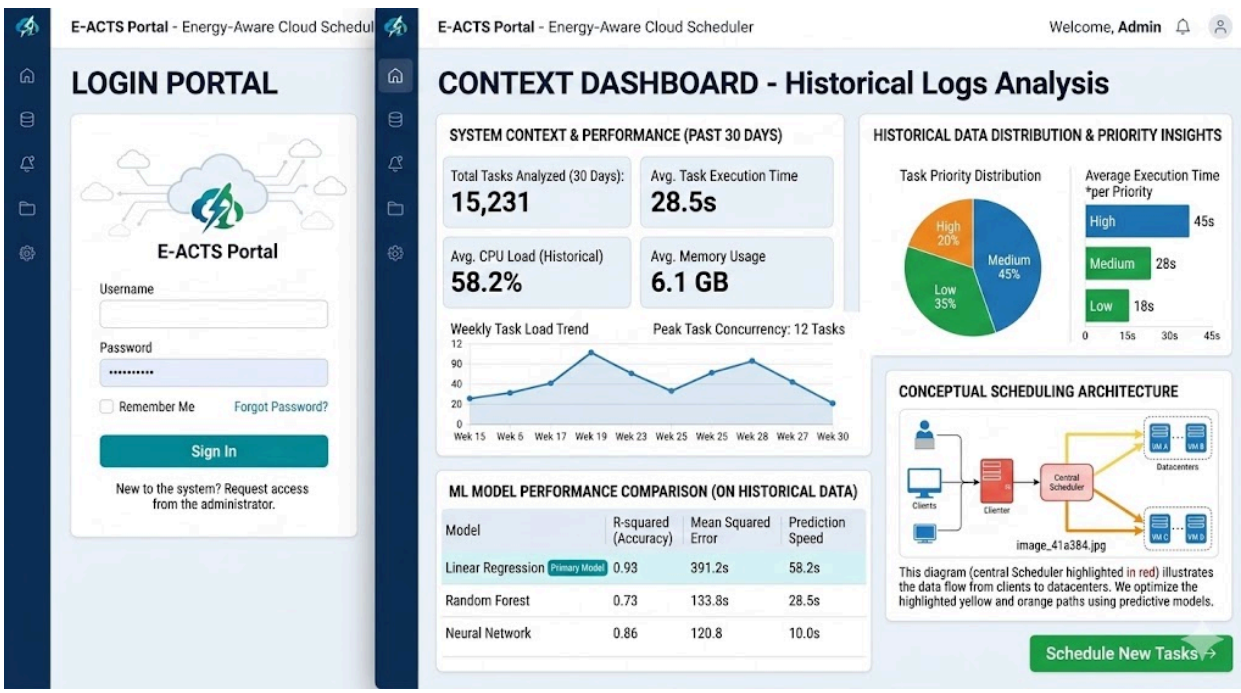
Login Page

- **What it is:** A simple, clean admin login screen.
- **Vibe:** Minimalist cloud infrastructure.

The Insights Dashboard (Welcome Page)

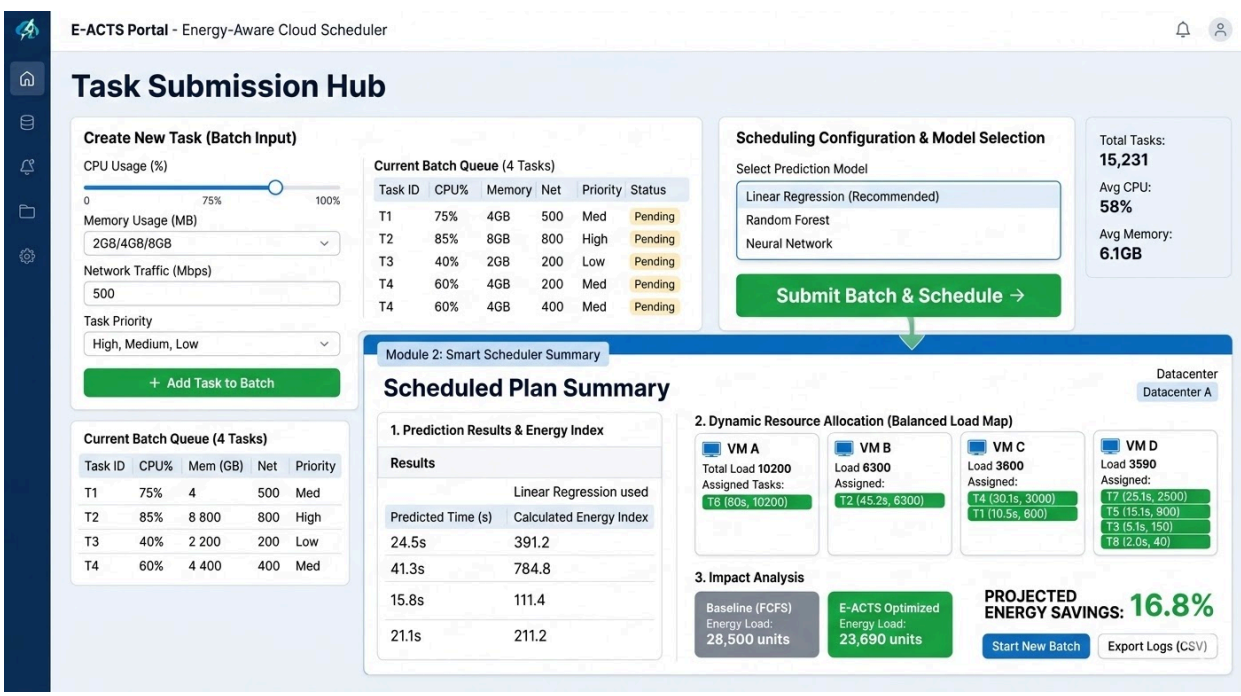
- **What it is:** This page shows the user what our ML models learned from.
- **Features:**
    - Read my `cloudscheduling.csv` file and show some cool summary stats on cards (e.g., Total Historical Tasks Analyzed, Average CPU Load, Average RAM).
    - Show a simple chart (like a pie chart of Task Priorities: High/Med/Low).
    - A big "Schedule New Tasks" button to move to the next page.



Task Submission Hub (The Input Page)

- **What it is:** Instead of one task at a time, the user needs to build a **Batch Queue**.
- **Features:**
    - **Left Side (Input Form):** Sliders or text boxes for the user to input a new task's parameters: CPU Usage (%), RAM (MB), Network IO (MB/s), and Priority (1, 2, or 3).
    - **"Add to Queue" Button:** Clicking this just saves it to a list on the screen, it doesn't send it to the backend yet.
    - **Right Side (The Queue):** A table showing the tasks the user just created.

- **Bottom:** A dropdown to select the ML Model (Linear Regression, Random Forest, NN) with Linear Regression set as the default.
- **"Submit Batch" Button:** This sends the whole table of tasks to the backend.

The Smart Scheduler Summary (The Results Page)

- **What it is:** This is where the magic happens. The backend predicts the times, calculates the energy, and balances the load. We display it visually.
- **Backend Logic required for this page:**
  - Run the submitted tasks through the selected ML model to get the `Predicted Execution Time` (in seconds).
  - **Calculate Energy Index:** For each task, use this formula: `Energy_Index = predicted_time * (task_cpu_usage + task_ram_usage)`.
  - **Load Balancing Algorithm:** Sort the tasks from highest `Energy_Index` to lowest. We will pretend we have 4 Virtual Machines (VM_A, VM_B, VM_C, VM_D). Loop through the sorted tasks and assign the next task to whichever VM currently has the *lowest* total Energy Index.
- **Frontend Display for this page:**
  - **Top:** A table showing the user's batch, plus the newly predicted Time and the calculated Energy Index.
  - **Middle (Visual Map):** 4 big boxes representing the 4 VMs. Show which Task IDs got assigned to which box based on your backend load balancing algorithm.
  - **Bottom:** A "Start Over" button.